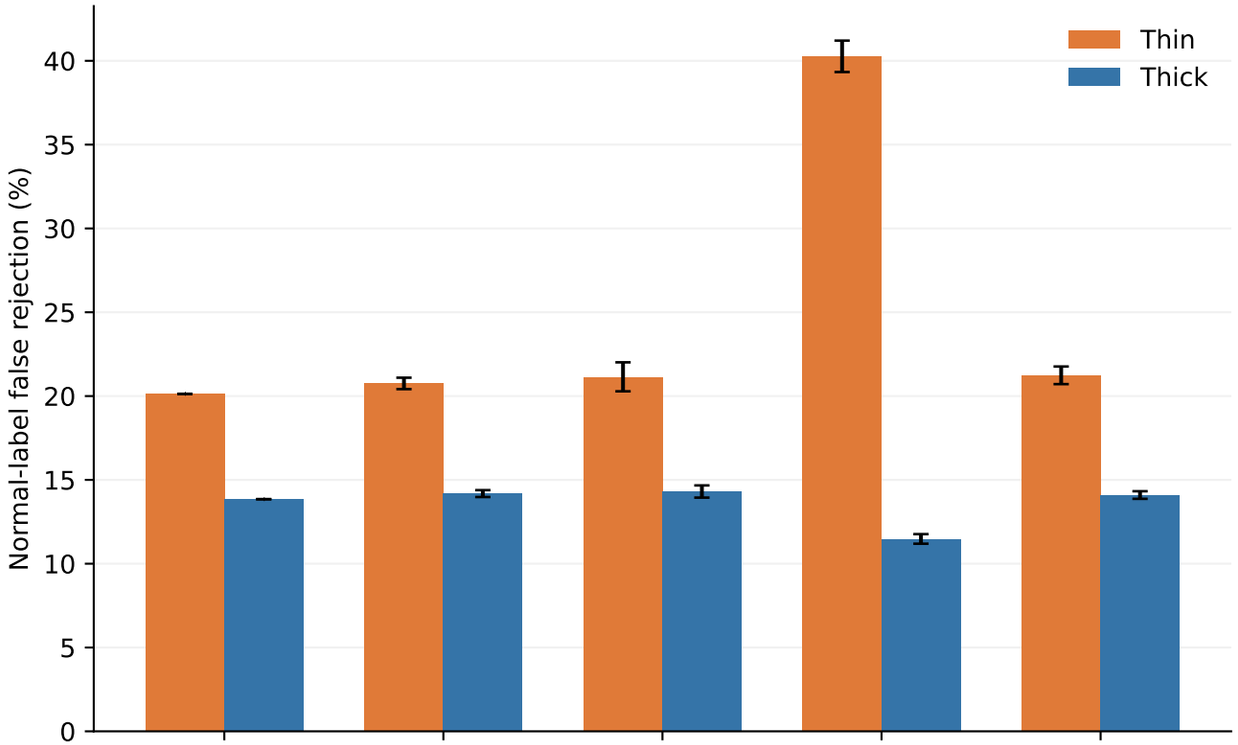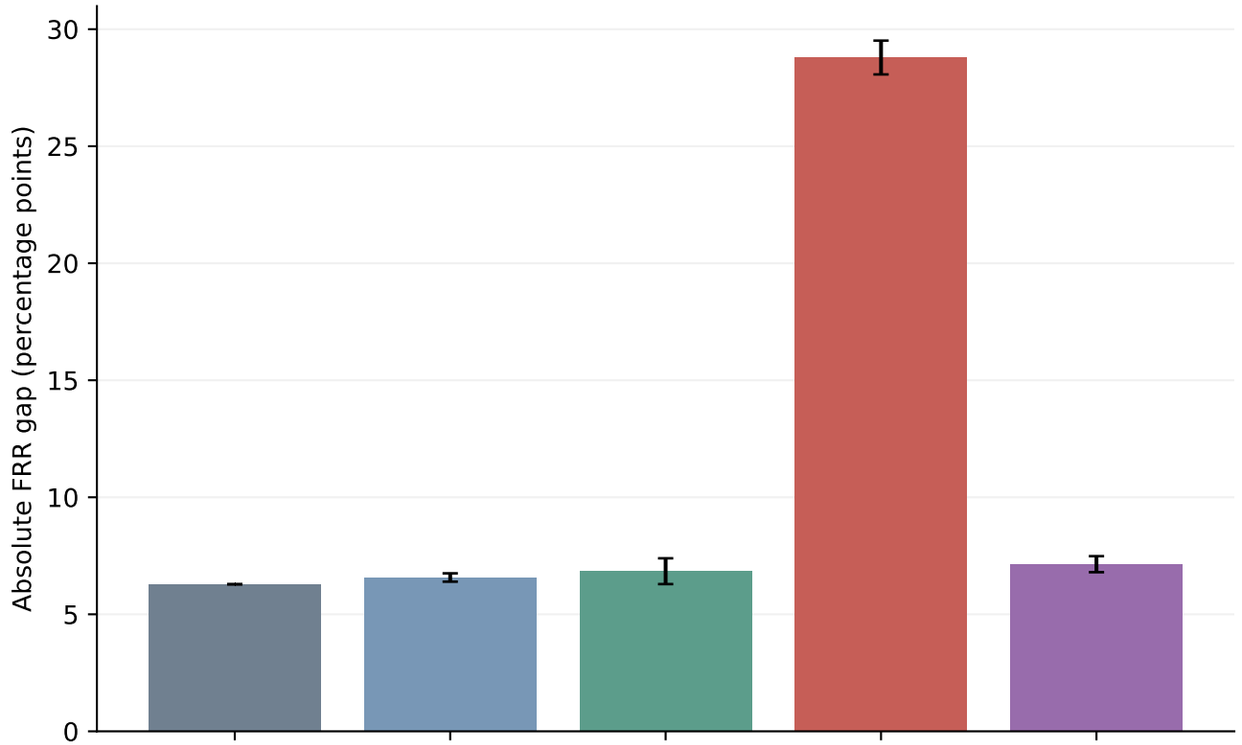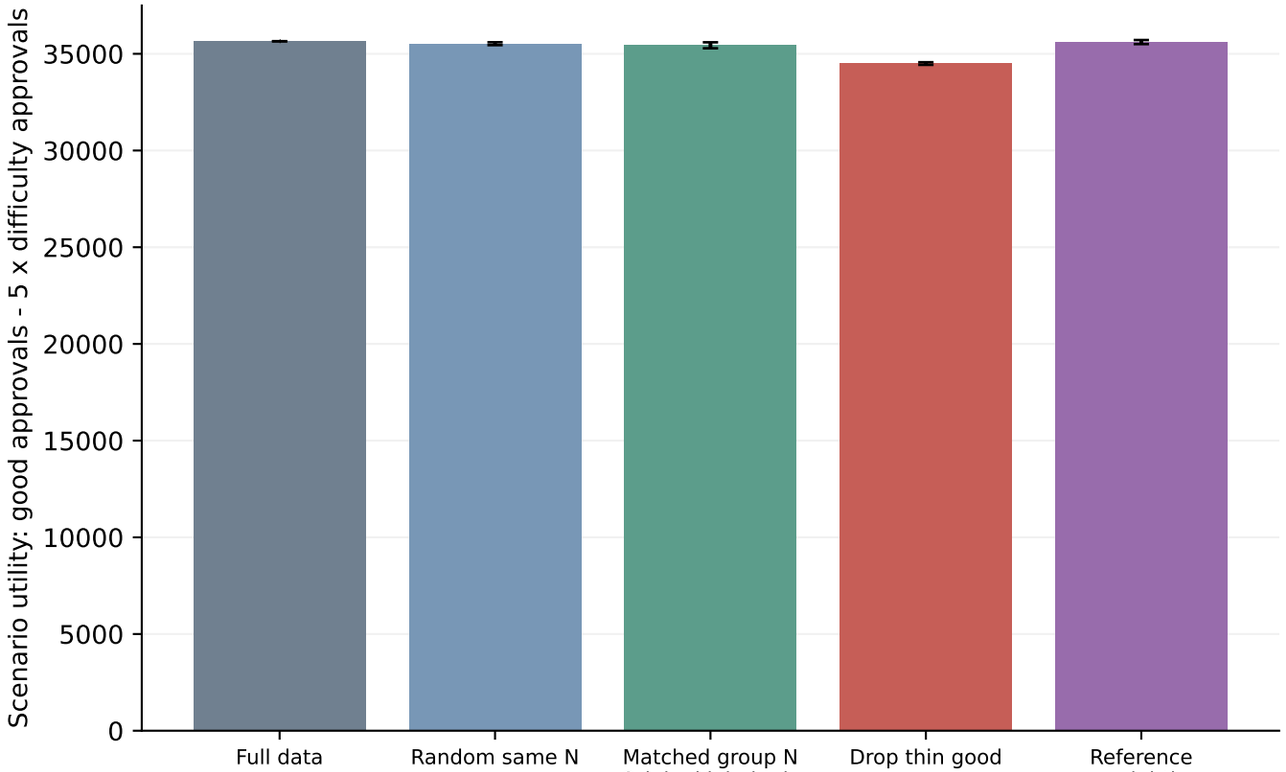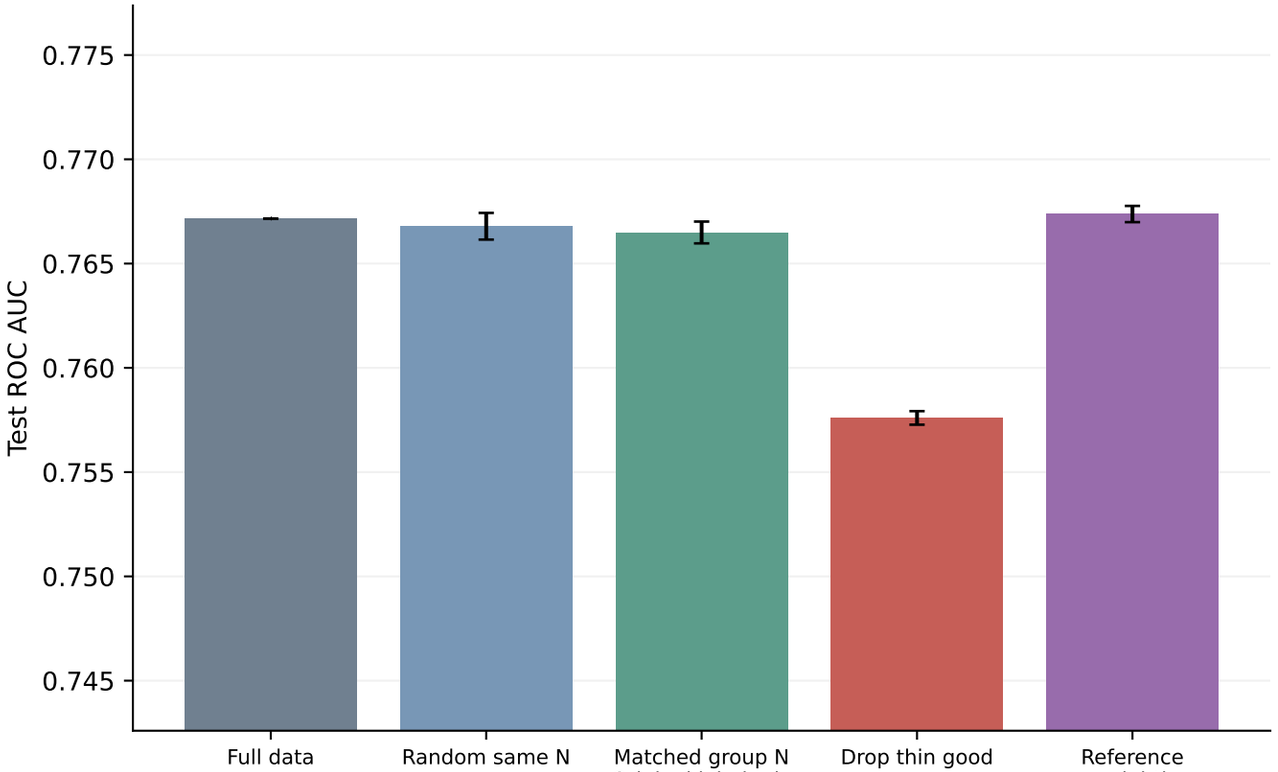Controlled sampling: data volume, label selection, and reference correction